

A Real-Time Edge Intelligence Framework for 6G-Enabled IIoT Manufacturing Systems

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Abstract

Modern smart manufacturing environments demand ultra-reliable, low-latency communication infrastructures to monitor high-frequency Industrial IoT (IIoT) telemetry without performance degradation. This paper presents an end-to-end edge intelligence framework that leverages **6G Ultra-Reliable Low-Latency Communication (URLLC)** metrics alongside physical machine parameters to predict manufacturing efficiency states in real time. By integrating advanced **Synthetic Minority Over-sampling Technique (SMOTE)** pipelines with optimized **XGBoost** classification engines, the proposed framework achieves an overall prediction accuracy of **99.7%**, effectively resolving the class imbalance inherent in industrial anomaly detection. Furthermore, the incorporation of **SHapley Additive exPlanations (SHAP)** breaks the “black-box” nature of machine learning, providing transparent root-cause analysis for shop-floor operators. The complete architecture is deployed via an interactive control room dashboard, bridging advanced predictive analytics with actionable industrial execution.

1 Introduction

The transition toward Industry 4.0 and subsequent autonomous smart manufacturing paradigms requires continuous, fine-grained monitoring of physical and network parameters. Traditional industrial monitoring systems rely heavily on reactive threshold checks, which fail to capture the multi-variable, non-linear interactions between network health and physical machine performance.

With the advent of **6G networks**, industrial automation benefits from massive device connectivity and deterministic sub-millisecond latency. However, fluctuating network characteristics such as jitter, packet loss, and high latency directly impact the stability of real-time control loops.

To bridge this gap, this research introduces a unified intelligence framework that:

1. Fuses high-frequency IIoT sensor telemetry (temperature, vibration, power consumption) with 6G network metrics (latency, packet loss).
2. Deploys an ensemble-ready gradient boosting classifier robust to skewed industrial datasets via advanced over-sampling.
3. Implements real-time explainable AI (XAI) to ensure deterministic root-cause interpretability for plant operators.

2 Methodology & System Architecture

2.1 Dataset Schema & Feature Engineering

The experimental evaluation utilizes a robust manufacturing telemetry dataset comprising **100,000 temporal records** across multiple machine nodes. Each record maps physical operational states against network performance vectors. To elevate raw telemetry into high-value predictive signals, domain-specific features were engineered:

- **Energy Efficiency Ratio:** Quantifies production output relative to power drawn, defined as:

$$\text{Energy_Efficiency} = \frac{\text{Production Speed}}{\text{Power Consumption} + \epsilon}$$

- **Error-to-Output Ratio:** Tracks production quality stability relative to unit velocity.
- **Network Reliability Score:** Captures composite 6G communication degradation via latency and packet loss weighting:

$$\text{Network_Reliability} = \text{Latency} \times \left(1 + \frac{\text{Packet Loss}}{100} \right)$$

- **Sensor Stress Index:** Evaluates mechanical fatigue through thermal-vibrational coupling:

$$\text{Stress_Index} = \text{Temperature} \times \text{Vibration}$$

2.2 Handling Class Imbalance via SMOTE

Industrial failure and medium-efficiency states typically occur sparsely compared to stable baseline operations, resulting in severe class imbalance. Training baseline models on raw data skews predictions toward majority classes, failing to capture critical degradation precursors.

To counteract this, the training partition was balanced using the **Synthetic Minority Over-sampling Technique (SMOTE)**, which interpolates new minority class instances along line segments joining k-nearest neighbors in the feature space. This intervention balanced the training distribution, preventing decision-boundary bias.

2.3 Predictive Modeling with XGBoost

An **Extreme Gradient Boosting (XGBoost)** classifier was selected for multi-class efficiency status prediction (**High**, **Medium**, **Low**). Configured with multi-class softmax objectives (`multi:softprob`), depth constraints (`max_depth = 6`), and controlled learning rates (`learning_rate = 0.1`), the model optimizes regularized objective functions to prevent overfitting on high-frequency industrial logs.

3 Results & Experimental Evaluation

The pipeline was executed on an 80/20 stratified train-test split (80,000 training samples, 20,000 test samples).

3.1 Baseline vs. Optimized Performance

A multi-class Logistic Regression baseline established an initial accuracy floor of **91.97%**, but struggled heavily with intermediate states, yielding an F1-score of **0.79** for the **Medium** efficiency class due to class skew.

Following SMOTE application and XGBoost training, the model performance experienced a dramatic uplift:

- **Overall Test Accuracy: 99.71%**
- **Macro-Averaged F1-Score: 0.99**

Table 1: Detailed Classification Metrics (XGBoost + SMOTE)

Efficiency Class	Precision	Recall	F1-Score	Support
High	1.00	0.99	0.99	625
Low	1.00	1.00	1.00	15,464
Medium	0.99	0.99	0.99	3,911
Accuracy / Macro Avg	1.00	0.99	0.99	20,000

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3.2 Explainable AI (SHAP) Integration

To satisfy industrial safety and auditability standards, **TreeExplainer** was integrated into the pipeline to compute exact Shapley values. This translates model outputs into explicit feature contributions, allowing operators viewing the live dashboard to instantly identify whether a sudden drop in efficiency is driven by physical anomalies (e.g., thermal spikes) or 6G network synchronization bottlenecks.

4 Conclusion & Future Scope

This research successfully demonstrates that combining 6G network metrics with IIoT physical telemetry through an optimized XGBoost-SMOTE pipeline yields near-perfect manufacturing efficiency classification (**99.7% accuracy**). By embedding real-time SHAP explainability within a cloud-deployed Streamlit control room, the framework bridges the gap between theoretical machine learning and mission-critical industrial execution.

Future iterations will focus on deploying this framework onto edge-computing gateways (such as NVIDIA Jetson modules) to minimize inference latency down to sub-millisecond thresholds directly on the factory floor.